Virtual University	**Assignment No. 01 [Non-Graded]** **SEMESTER Fall 2013** **CS301- Data Structures**	**Total Marks: 30** **Due Date: 18/11/2013**

Instructions

Please read the following instructions carefully before solving & submitting assignment:

It should be clear that your assignment will not get any credit (zero marks) if:

- **The assignment is submitted after due date.**
- **The submitted code does NOT compile.**
- **The submitted assignment file is not in .CPP format.**
- **The submitted assignment file does not open or corrupted.**
- **The assignment is copied (from other student or ditto copy from handouts or internet).**

Uploading instructions

For clarity and simplicity, You are required to Upload/Submit only ONE zip file which will contain solutions of both questions.

- **Don't wait for grace day. Grace Day is given only if there is problem with LMS on due date. Submit your solution within due date.**
- **Note that no assignment will be accepted through email if there is any problem on grace day.**

Note: Use only dev-C++ IDE.

This assignment is non-graded; marks of this assignment will not be included in your course final grades.

Objective

The objective of this assignment is

- To make you familiar with working of linked list and stack data structures and programming techniques to implement and understand working of these data structures.

For any query about the assignment, contact at cs301@vu.edu.pk

GOOD LUCK

	Marks: 30

Question 1: **Marks 10**

Write a C++ program to.

1) Take as input any integer value from user.

2) Generate all the prime numbers up to that integer value and display them on screen.

3) Now create a linked list by using those prime numbers generated above and display the linked list in sorted ascending order.

Sample output screen shot of program is given below

```
Enter Number
40
Prime numbers between 1 to 40 are:

2 3 5 7 11 13 17 19 23 29 31 37
 List elements are given below

 Element 2

 Element 3

 Element 5

 Element 7

 Element 11

 Element 13

 Element 17

 Element 19

 Element 23

 Element 29

 Element 31

 Element 37
Press any key to continue . . .
```

Note: In the above screen shot, program asks the user to enter any number. User enters 40. Program then displays prime numbers from 1 to 40. These prime numbers are then inserted in a linked list and these prime numbers from linked list are then displayed in sorted ascending order.

Solution Guidelines:

1. First understand the code given in handouts about link list.
2. Remember numbers that are only dividable by itself and 1 are called prime numbers.
3. Find a prime number display it and inset into linked list.
4. Finally display all number in list sorted order.

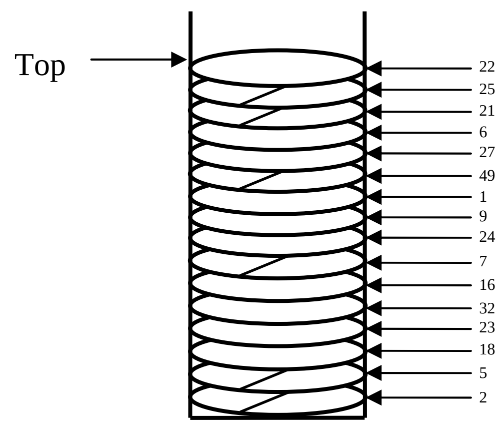
Question 2: **Marks 20**

Write a C++ program to implement stack of marble plates. Your program should cover the following scenario.

Suppose a waiter in a hotel is collecting plates from different tables and giving to dish washer. Dish washer is washing plates and throwing them into stack of plates. Some plates hit others very hard to break other (one below it) and/or itself.

You need to develop an application that will find how many plates are broken and plates from which tables are broken.

You will identify plates with unique numeric identity number. If plate's identity number will fully dividable by 5, will break itself and one plate below it. While the number fully dividable by 7 will break only itself.



Program's sample output is given below

```
Set size of stack: 10
How many values you want to enter into stack: 11

Enter plate identity number that you want to insert into stack: 25

Enter plate identity number that you want to insert into stack: 24

Enter plate identity number that you want to insert into stack: 21

Enter plate identity number that you want to insert into stack: 18

Enter plate identity number that you want to insert into stack: 19

Enter plate identity number that you want to insert into stack: 15

Enter plate identity number that you want to insert into stack: 14

Enter plate identity number that you want to insert into stack: 9

Enter plate identity number that you want to insert into stack: 5

Enter plate identity number that you want to insert into stack: 7

Enter plate identity number that you want to insert into stack: 2

 Unable to insert 2. Stack is full, can't insert more

Following plates are broken

The plate with identity number 7 is broken

The plate with identity number 5 is broken

The plate with identity number 9 is broken

The plate with identity number 14 is broken

The plate with identity number 15 is broken

The plate with identity number 19 is broken

The plate with identity number 21 is broken

The plate with identity number 25 is broken

Stack is empty, sorry can't pop more

Total 8 plates are broken
```

Solution Guidelines:

1. First understand the code given in handouts about stack.
2. Get stack size from user to allocate space for stack dynamically.
3. Get identity number of plates that user want to enter into stack.
4. Get value one by one and push into stack.
5. While popping the values check what value is and show if it is broken.
6. Count all broken plates after popping all.
7. Don't allow popping if stack is empty and pushing if stack is full.

Lectures Covered: This assignment covers Lecture # 1-6
Deadline: Your assignment must be uploaded / submitted on / before, **18 November, 2013.**